University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Digital Logic Design	**Course Code:** CMPE-222L
Assignment Type: Lab	**Dated:** 20th October 2026
Semester: 3rd	**Session:** 2025
Lab Number: 4	**CLOs to be covered:** CLO 1, CLO 2, CLO 3
Lab Title: Introduction to gate level design of combinational circuits.	**Teacher Name:** Fareed Jafri

Lab Goals

- Electronically add or subtract 4-bit numbers through the 2's complement method (using XOR gates).
- Design and implement a simplified circuit in SOP form (using AND OR and NOT gates).

Equipment Required

- **ICs:** 7408 (1x), 7432 (1x), IC 7483 (1x), IC 7486 (1x)
- **Tools:** IC puller (1x), Nose pliers (1x), Wire stripper (1x)

Lab Evaluation

	CLOs	Maximum marks	Marks obtained
Task 1a	1	3	
Task 1b	3	2	
Task 2a	2	2	
Task 2b	1	3	
Total			**/10**

Rubrics for Current Lab Evaluation

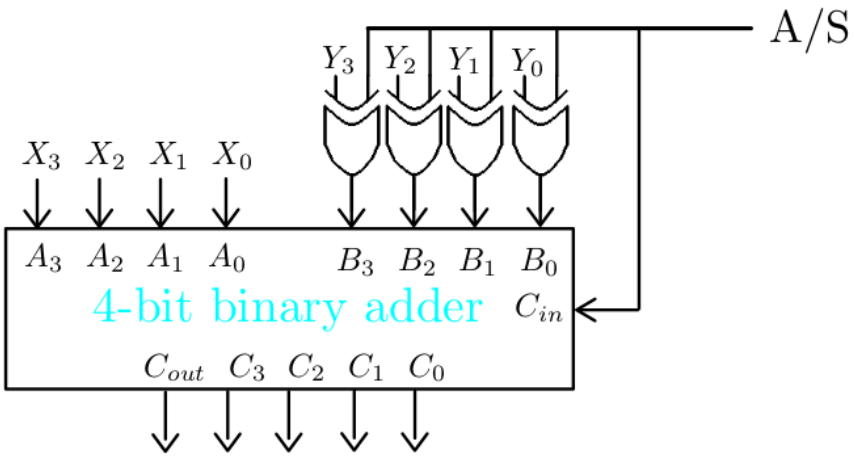
Scale	Marks	Level	Rubric	CLOs
Completed	**100%**	L1	Completed the task correctly.	All
Great attempt	**75%**	L2	Almost completed the task, with minor deficiencies	2,3
Good attempt	**50%**	L3	Did not complete the task but managed to reasonably attempt it.	All
Basic attempt	**25%**	L4	Did not complete the task but managed to basically attempt it.	All
Did not attempt	**0%**	L5	Did not complete the task or attempt to do it.	All

Electronically add or subtract 4-bit numbers by implementing the 2's complement method

Task 1a [CLO1, 3 Marks]

The figure below shows a schematic for a module which can add or subtract two 4-bit signed integers X = $X_3X_2X_1X_0$ and Y = $Y_3Y_2Y_1Y_0$ to produce a 4-bit signed integer output C = $C_3C_2C_1C_0$. The output C = X+ Y when A/S is 0, and C = X – Y when A/S is 1 (as long as signed overflow does not occur).

The pin out of the 7483 and 7486 ICs is shown in the 'Pin out' section at the end. Implement the circuit below using these ICs. Switches S0 to S3 provide the first 4-bit input X. Switches S4 to S7 provide the second 4-bit input Y. Switch S8 controls the A/S input. LEDs L0 to L3 record the 4-bit output C, of the module.



Task 1b [CLO3, 2 Marks]

Now fill in the following table using the circuit from Task 1a. Put a circle on those entries of C where signed overflow has occurred. All entries are in binary.

X	Y	A/S	C		X	Y	A/S	C
0000	0101	0			0110	1000	1	
0001	0101	0			0110	1001	1	
0010	0101	0			0110	1010	1	
0011	0101	0			0110	1011	1	
0100	0101	0			0110	1100	1	
0101	0101	0			0110	1101	1	
0110	0101	0			0110	1110	1	
0111	0101	0			0110	1111	1	

Design and implement a simplified AND-OR-NOT circuit in SOP form

Task 2a [CLO2, 2 Marks]

The truth table for a combinational circuit (with a 3-bit input xyz and single bit output F) is shown below.

x	**y**	**z**	**F**
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

1. Design an AND-OR-NOT circuit in Σ form from truth table. (Only write the boolean expression for the function in Σ form, **without drawing the logic diagram**).

2. Using the **rules of boolean algebra** simplify this combinational circuit in SOP form. Show your working here.

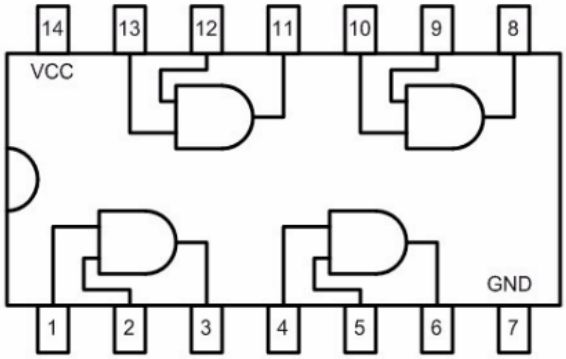
3. Draw the logic diagram for your simplified expression.

Task 2b [CLO1, 3 Marks]

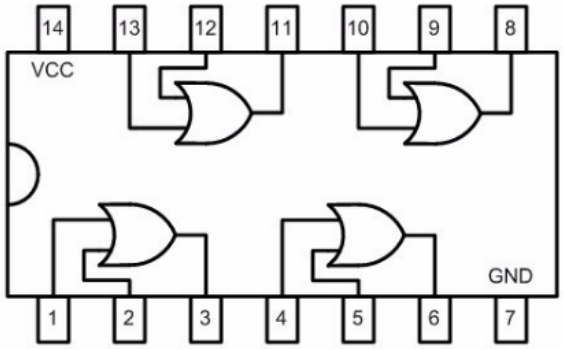
The pin out of the 7408 and 7432 ICs is shown in the 'Pin out' section at the end. Use these ICs to implement your logic diagram in Task 2a and show that it follows the truth table given. Switches S7, S8 and S9 control the inputs x,y and z. Connect the output F to any LED of your choice. Wire up the circuit and show that it works correctly.

Pin out

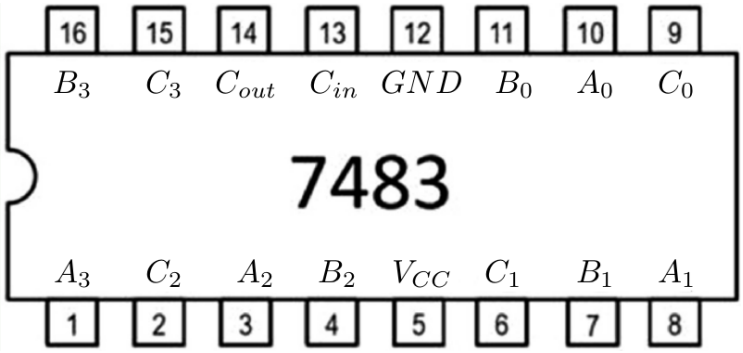
7408 IC



7432 IC



7483 IC



7486 IC

